

GDG WINDSOR · BUILD WITH AI · EDGE / ON-DEVICE TRACK

Gemma Monsters

An on-device math adventure for the Grade 9 EQAO.

Five monsters guard the Citadel. Each one is a trick — a wrong idea that feels right.

Defeat the idea, and the golden gate opens. Nothing ever leaves the laptop.

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The problem: wrong answers are symptoms, not causes

- Every Ontario Grade 9 student writes the EQAO math assessment. Many arrive with **specific, well-documented tricks** — wrong ideas that feel right, not general weakness.
- A student who answers $\frac{2}{3} + \frac{1}{4} = \frac{3}{7}$ doesn't need more practice. They need someone to notice they **add tops and bottoms straight across** — and fix exactly that.
- Teachers can't do this per-student, per-trick. **So we turned each trick into a monster you can fight.**

Calculate $\frac{2}{3} + \frac{1}{4}$

A) $\frac{11}{12}$

correct

B) $\frac{3}{7}$

adds straight across

C) $\frac{3}{12}$

didn't rescale tops

D) $\frac{11}{7}$

mixed method

Every wrong option is the exact result of a *known* trick — so the wrong answer itself is the diagnosis.

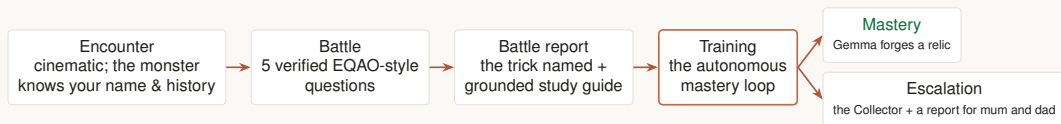
The game world: the Citadel

A gothic castle nexus in three.js — drag to orbit. Five monsters idle on floating platforms, each guarding one curriculum strand and a sealed **golden gate**. Someone is locked in the keep; defeat all five to open it.

Monster	Strand (MTH1W)	The trick it feeds on
Fractis	Number	adding fractions straight across, tops and bottoms
Equazor	Algebra	signs that slip when balancing an equation
Statiq	Data	mean and median blurred into the same idea
Polygor	Geometry & Measurement	formulas that are <i>almost</i> right
Ledgerling	Financial Literacy	interest skimmed off a sloppy budget

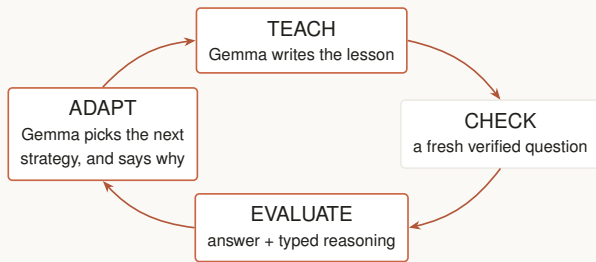
- The game captures your **hero name**; monsters address you by it in full-screen encounter cinematics.
- And they **remember**: Gemma writes each monster's greeting from your actual battle history.

The player journey



- Miss a question and the agent reads the wrong option's tag, names the **exact trick**, and builds a study guide grounded in the verified solution — with real exponent and fraction notation on screen.
- Then it **pursues a goal**: keep teaching, with different strategies, until you *demonstrate* the trick is gone — or the Collector comes for you.

An agent, not a chatbot



The reasoning gate: a right answer with shaky typed reasoning **does not count** — Gemma grades the explanation (closed labels), not just the bubble.

Explainable everywhere: every decision shows its why — “Trying a visual walkthrough — you said you can’t see why the denominators matter.”

It cannot run away: 4-attempt cap, 12-model-call budget, exhaustion → clean hand-off.

Orange boxes = Gemma. Every session provably terminates.

Where Gemma does the thinking

rubric: Gemma Integration, 30%

Gemma's job (8 call sites)	Guardrail
Write the lesson & study guide	grounded in the verified solution; explains <i>why</i> , never recomputes
Grade typed reasoning	closed labels; fail-open — never punishes the student
Choose the next strategy	limited to the remaining ladder; states its rationale; code fallback
Author fresh check questions	bank-first; must pass a blind self-solve audit or is discarded
Monster memory lines	one line, from real battle history; scripted fallback
Forge the mastery relic	flavour only — the facts are injected by code
Write the mum-and-dad report	narrative only; every number comes from the session log
Coach the speed trials	reads miss + latency telemetry; names the pattern; sets a 3-question drill

- Remove Gemma and there is no game. **But the math truth never depends on it.**
- One auditable door (`gemma_client.py`); model size is a dial: `gemma3:12b` primary, `gemma3:1b` light.

The Collector, his lieutenants, and the Gemma coach

THE COLLECTOR — a giant skull boss, summoned when the mastery loop escalates: a mental-math speed trial, 3 lives, he attacks on every miss. Gemma writes the report for mum and dad alongside — the agent knows its limits.

Three lieutenants, one mental-math lane each:

- **Twinfang** — doubles and halves
- **The Niner** — the nines shortcut
- **Splitjaw** — make-a-ten

90-second war-clock skirmishes with streak escalation; every miss and its latency logged.

GET COACHED BY GEMMA

- Before the fight, Gemma **whispers the lane's mental strategy** in its own words.
- After it, Gemma reads the miss + latency telemetry, **names your miss pattern**, teaches the fix, and sets a **3-question drill**.
- The battle is pure client-side speed; **the coaching is where the model earns its keep**.

The whole wing runs to a procedural WebAudio soundscape — no audio files shipped.

We caught our model failing — and engineered around it

Every guardrail exists because live testing broke something. The commit history is the proof of work.

Caught in testing	Architectural fix
Generated a question with a wrong answer key	blind self-solve audit; bank-first check questions
Wrote a check question in Spanish	constrained generation prompts
Invented wrong arithmetic in an explanation ($\frac{2}{3} + \frac{1}{4} = \frac{17}{12}$)	grounding rule: explain <i>why</i> , never recompute; numbers only from the verified solution
Rambled when reconstructing multi-step errors	diagnosis by ground-truth tag lookup, not model reasoning
Emitted 2^3 and $2/3$ as raw markup	notation layer: real exponents and stacked fractions on screen

- The result: **the model is never the source of truth for math** anywhere a student can see it.

- Gemma runs locally via **Ollama**:
gemma3:12b primary, gemma3:1b light — one env var. Works in airplane mode.
- The 3D world is local too: **three.js** and the **CC0 Quaternius monsters** are vendored — no CDN.
- The soundscape is **synthesized procedurally in WebAudio** — zero audio files.
- A student's struggles are a protected record — **they stay on their machine.**

Companion Kaggle notebook: Gemma 4 12b-it on a cloud GPU — the app never leaves the laptop.

The student's laptop
three.js Citadel + Streamlit
battles · agent loop
· coach · reports
↕
Ollama + Gemma
12B primary · 1B one flag away

nothing crosses this line
(the internet)

The 90-second live demo

1. **The Citadel** — drag the castle; five monsters idle on their platforms; the keep is sealed.
2. **Encounter** — Fractis addresses us by our hero name and *remembers our last visit*.
3. **The miss** — we add $\frac{2}{3} + \frac{1}{4}$ straight across; the battle report names the exact trick.
4. **The agent adapts** — we answer wrong *and type our flawed reasoning*; the agent switches strategy **and explains why**, live on screen.
5. **The relic** — two fresh correct answers with solid reasoning → Gemma forges our trophy.
6. **The Collector's wing** — a 90-second lieutenant skirmish, then **GET COACHED BY GEMMA**: it reads our misses, names the pattern, and drills the fix.

Everything shown runs live on this laptop. Wi-Fi can be off.

- **Local tool use** — Gemma emits constrained JSON tool calls; a local SymPy verifier proves every answer deterministically. (tool use for a model that doesn't natively support it)
- **Beyond the golden gate** — the rescue arc continues: new keeps, new boss lanes, harder relics — same agent brain underneath.
- **Classroom mode** — session records roll up across a whole class, still local-first: the school's data stays in the school.

HOW WE SCORE OURSELVES ON THE RUBRIC

Gemma Integration 30%

8 roles, all guardrailed

Innovation 30%

a tutor agent you can fight

Functionality 20%

live, on-device, terminates

Writeup 20%

honest engineering story

Every monster is a trick —
a wrong idea that feels right.

Defeat the idea, and the gate opens.

A student walks in adding fractions straight across. Twenty minutes later they've *proven* the trick is gone — and nothing about them ever left their laptop.

github.com/EdTechDL/gemma-without-borders · live demo + clonable notebook

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